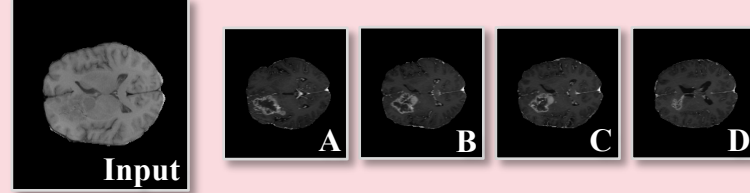


All VLM Supervision is constructed Exclusively From Paired Translation Samples

1 Task 1: Conditional Target Selection

Select the image that is the **one-to-one counterpart** of the reference in the **target modality**?

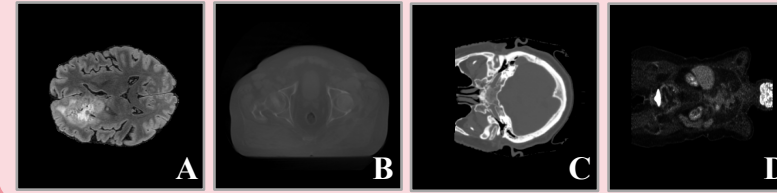


: Option B

Forces slice-level correspondence & content invariance

2 Task 2: Modality Identification

Provide the **modality** for each panel A–D?

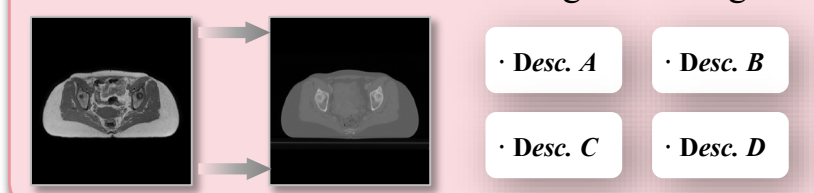


: "A" : "FLAIR (MRI)", "B" : "CBCT",
"C" : "CT", "D" : "PET"

Explicitly learns modality as a controllable condition

3 Task 3: Transformation Instruction Alignment

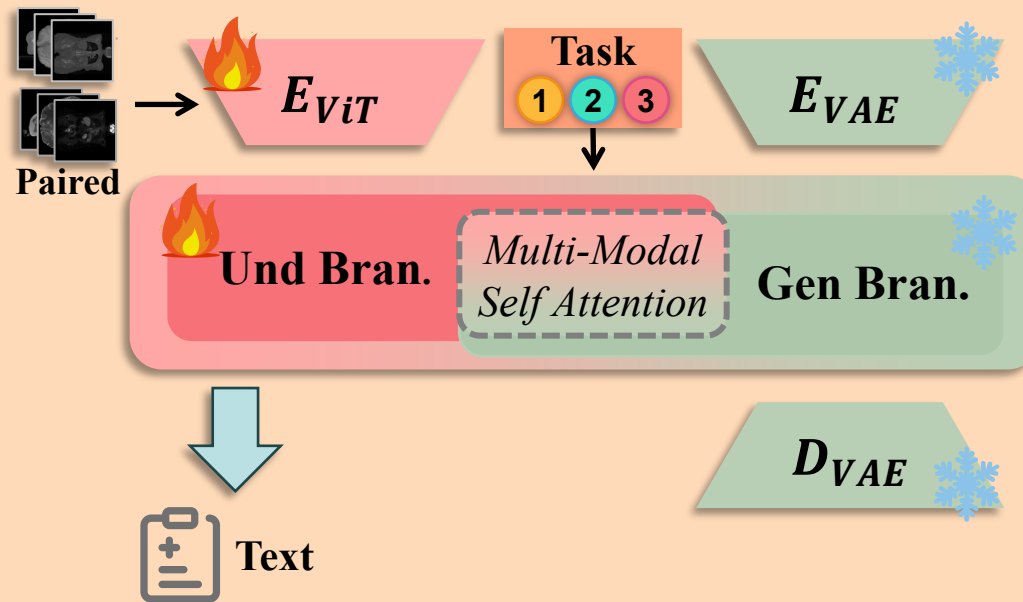
Read the candidate instructions and select the one that **BEST describes** image 1 → image 2.



: Description A

Anchors translation direction through language-aligned semantics

Stage I: Generation-Grounded Understanding



Stage II: Understanding-Conditioned Generation

